

Spring 2025

Math F251X

Calculus 1: Midterm 2

Name: Solomon

Section: ☐ 9:15am (James Gossell)
☐ 11:45am (Kevin Meek)
☐ async (Deven Barnett)

Rules:

- Partial credit will be awarded, but you must show your work.
- You may have a single handwritten $3'' \times 5''$ notecard, both sides.
- Calculators are **not** allowed.
- Place a box around your FINAL ANSWER to each question where appropriate.
- Turn off anything that might go beep during the exam.

Good luck!

Problem	Possible	Score
1	10	
2	6	
3	12	
4	12	
5	12	
6	12	
7	10	
8	12	
9	8	
10	6	
Extra Credit	(5)	
Total	100	

11:50
12:45

1. (10 points)

The radius of a sphere decreases at a rate of 2 meters per second. Find the rate at which the volume decreases when the radius is 5 meters. The volume of a sphere is given by $V = \frac{4}{3}\pi r^3$.

$$V = \frac{4}{3}\pi r^3 \quad \frac{dr}{dt} = -2 \text{ m/s} \quad r = 5 \text{ m}$$

$$\frac{dV}{dt} = \frac{4}{3}\pi 3r^2 \frac{dr}{dt}$$

$$= \frac{4}{3}\pi 3(5)^2 (-2 \text{ m/s})$$

$$= 4(25)(-2)\pi \text{ m}^3/\text{s}$$

$$= -200\pi \text{ m}^3/\text{s}$$

Volume is decreasing at $200\pi \text{ m}^3/\text{s}$
when the radius is 5 m.

2. (6 points)

Let $g(x) = x + \cos(x)$. Determine the absolute maximum and absolute minimum of $g(x)$ on the interval $[0, 2\pi]$.

$$g'(x) = 1 - \sin(x)$$

$$0 = 1 - \sin(x)$$

$$1 = \sin(x)$$

$$x = \pi/2 \text{ on } [0, 2\pi]$$

$$g(0) = 0 + 1 = 1$$

$$g(\pi/2) = \pi/2 + 0 = \pi/2$$

$$g(2\pi) = 2\pi + 1 = 1 + 2\pi$$

$$\begin{aligned} \text{maximum value of } g(x): & \frac{1+2\pi \text{ at } x=2\pi}{1 \text{ at } x=0} \\ \text{minimum value of } g(x): & \end{aligned}$$

3. (12 points)

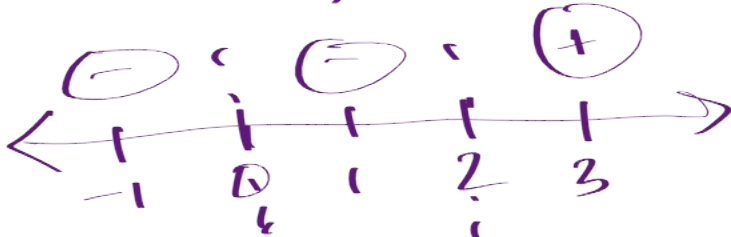
Let $g(x) = x^{4/3} - 8x^{1/3}$. Note that we have the following:

$$g'(x) = \frac{4x - 8}{3x^{2/3}} \quad g''(x) = \frac{4x + 16}{9x^{5/3}}$$

You must show your work for all parts to receive credit!

- a. Determine the intervals where g is **increasing** and where g is **decreasing**.

CP: $x=0, 2$ from $g'(x)=0$ or undefined



Increasing: $(2, \infty)$
Decreasing: $(-\infty, 0) \cup (0, 2)$

- b. Find the x values where any **local maxima** occur and where any **local minima** occur.

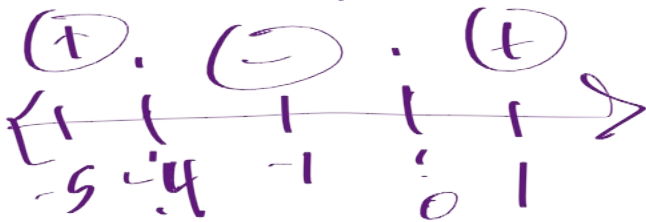
Investigate $x=2$
 $g''(2) > 0$, concave up

so $x=2$ is a local minima

$x=0$ is neither since $g'(x) < 0$ on either side of $x=0$

- c. Find the intervals where g is **concave up** and where g is **concave down**.

Critical points for $g''(x)$ are $x=0, -4$



Concave up: $(-\infty, -4) \cup (0, \infty)$
Concave down: $(-4, 0)$

- d. Find the x values where any **inflection points** occur.

Inflection points at $x=0, -4$
defined for $g(x)$ and concavity changes.

4. (12 points)

Evaluate the following limits. **Show your work**, including appropriate use of limit notation. If you use L'Hôpital's rule, you must indicate where you are using it by writing $\frac{H}{L}$ or $\frac{L'H}{L'H}$ or something similar. Use ∞ or $-\infty$ where appropriate, and if the limit does not exist, write DNE and provide a justification.

a. $\lim_{t \rightarrow \infty} \frac{2x^2 - 3x}{12 - 5x^2}$ form $\frac{\infty}{\infty}$ can divide by x^2

$$= \lim_{t \rightarrow \infty} \frac{2 - \frac{3}{x}}{\frac{12}{x^2} - 5} = \boxed{-2/5}$$

b. $\lim_{\theta \rightarrow 0} \frac{4 \cos(\theta) - 4}{e^\theta - \theta - 1}$ gives $\frac{4-4}{1-0-1} = \frac{0}{0}$

(H) $= \lim_{\theta \rightarrow 0} \frac{-4 \sin \theta}{e^\theta - 1}$ form $\frac{0}{0}$

(H) $= \lim_{\theta \rightarrow 0} \frac{-4 \cos \theta}{e^\theta} = \frac{-4}{1} = \boxed{-4}$

c. $\lim_{x \rightarrow \infty} \frac{\sqrt{x}}{\ln(x^2)}$

form $\frac{\infty}{\infty}$
(H) $= \lim_{x \rightarrow \infty} \frac{\frac{1}{2} x^{-1/2}}{\frac{1}{2} 2x} = \lim_{x \rightarrow \infty} \frac{x^{1/2}}{4} = \boxed{\infty}$

d. $\lim_{x \rightarrow 0} x \cot x$

$\cot x = \frac{\cos x}{\sin x}$
form $0 \cdot \infty$

$= \lim_{x \rightarrow 0} \frac{x}{\sin x}$ form $\frac{0}{0}$

(H) $= \lim_{x \rightarrow 0} \frac{1}{\cos^2 x} = \boxed{1}$

5. (12 points)

We want to minimize the surface area of a closed cylinder (a can, a jar, a barrel, etc.) with volume 36π cm³. The surface area of a closed cylinder is given by $S = 2\pi r^2 + 2\pi rh$ and the volume of a cylinder is given by $V = \pi r^2 h$.

- a. Write the equation to be minimized in terms of one variable.

$$\begin{aligned} V &= \pi r^2 h \\ 36\pi &= \pi r^2 h \\ h &= \frac{36\pi}{\pi r^2} = \frac{36}{r^2} \end{aligned}$$

$$S = 2\pi r^2 + 2\pi r \left(\frac{36}{r^2} \right)$$

$$S = 2\pi r^2 + 72\pi r^{-1}$$

- b. Find the dimensions of the cylinder with the smallest surface area.

$$S' = 4\pi r - 72\pi r^{-2} \stackrel{\text{set}}{=} 0$$

$$4\pi r^3 = 72\pi$$

$$r^3 = 72\pi / 4\pi = 18$$

$$r = \sqrt[3]{18} \text{ cm}, h = \frac{36}{18^{2/3}} \text{ cm}$$

- c. Demonstrate that you have found the dimensions for the *minimum* surface area.

$$S'' = 4\pi + 144\pi r^{-3}$$

$$4\pi + \frac{144\pi}{(\sqrt[3]{18})^3} > 0, \text{ concave up.}$$

So $r = \sqrt[3]{18}$ cm is the minimum.

6. (12 points)

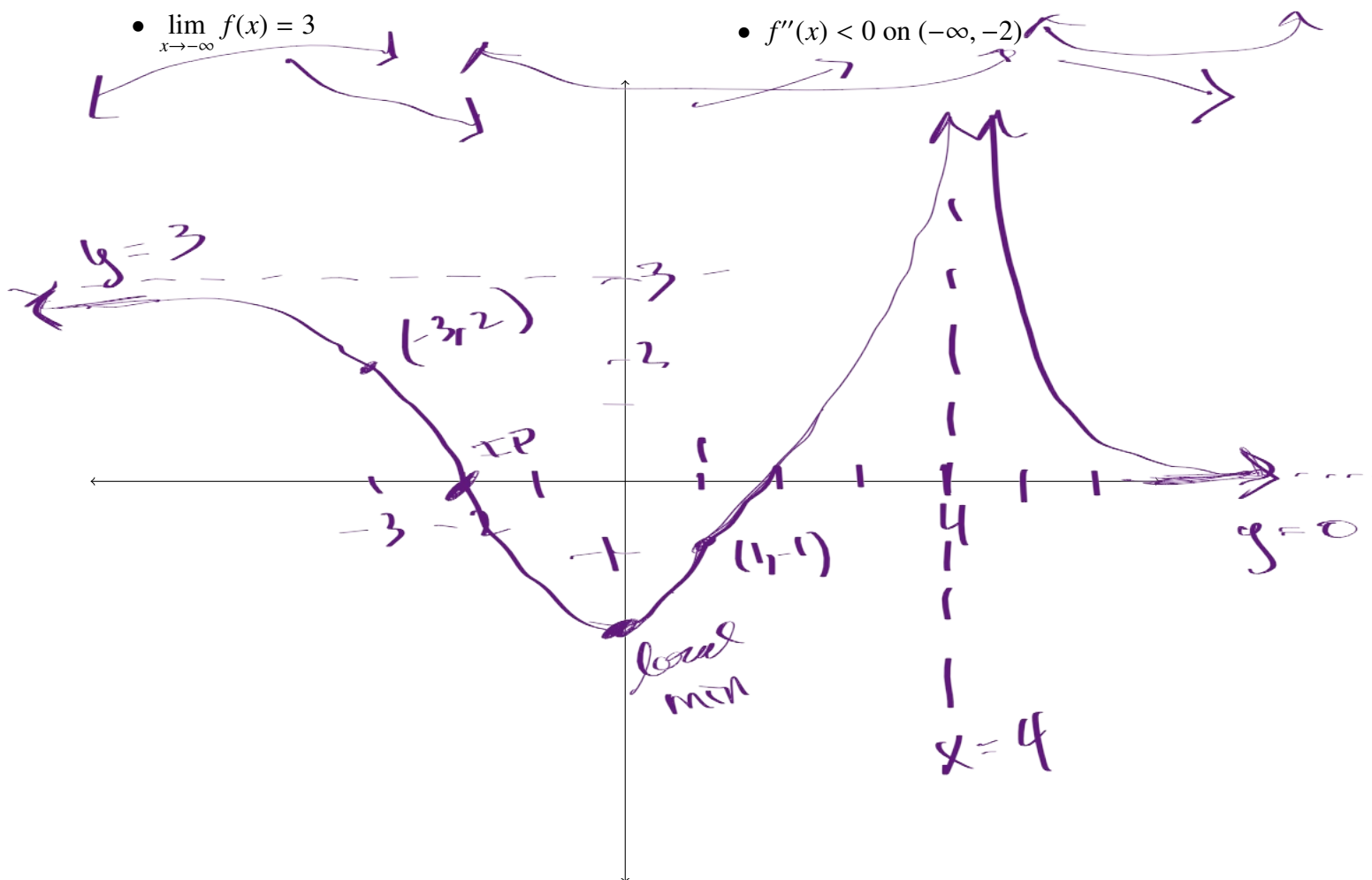
Sketch a graph of a function $f(x)$ that satisfies all of the following properties.

After drawing the graph:

- **Label** on the graph the following things, if they exist, by drawing a point on the graph and labeling: any local maximums by writing LOCAL MAX, local minimums by writing LOCAL MIN, inflection points by writing IP
- Draw any horizontal and vertical asymptotes with dashed lines and **label** them with their equation.
- Mark any important x -values and y -values on the x - and y -axes.

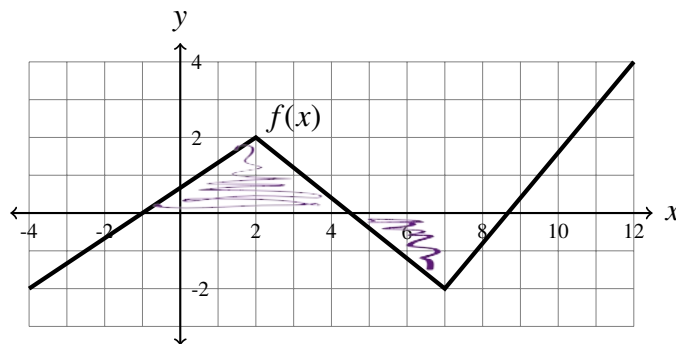
Properties:

- Domain is $(-\infty, 4) \cup (4, \infty)$
- $f(1) = -1$, $f(-3) = 2$
- $\lim_{x \rightarrow \infty} f(x) = 0$
- $\lim_{x \rightarrow -\infty} f(x) = 3$
- $f'(x) > 0$ on $(0, 4)$
- $f'(x) < 0$ on $(-\infty, 0) \cup (4, \infty)$
- $f''(x) > 0$ on $(-2, 4) \cup (4, \infty)$
- $f''(x) < 0$ on $(-\infty, -2)$



7. (10 points)

Consider the graph of the function $f(x)$ shown below:



- a. Determine $\int_{-1}^7 f(x) dx$. Show some work or say something about what you computed.

Area of triangles
 $= (5\frac{1}{2})(2)\frac{1}{2} + (2\frac{1}{2})(2)(\frac{1}{2})$
 $= 7.5$

- b. Determine $\int_{-1}^7 (2f(x) + 3) dx$. (Hint: Use part (a) above.)

$$\begin{aligned}
 &= 2 \int_{-1}^7 f(x) dx + \int_{-1}^7 3 dx \\
 &= 2(7.5) + 3(7 - (-1)) \\
 &= 15 + 24 \\
 &= 39
 \end{aligned}$$

8. (12 points)

Compute the following **indefinite integrals** (antiderivative). Give the most general answer, and show your work. Clearly indicate any substitutions you use in such a way that someone else can follow your work.

a. $\int \sqrt[5]{x^3} + 3e^x - \cos(x) dx$

$$= \frac{5}{8} x^{8/5} + 3e^x - \sin x + C$$

b. $\int x^2(3x+4) dx = \int 3x^3 + 4x^2 dx$

$$= \frac{3}{4} x^4 + \frac{4}{3} x^3 + C$$

c. $\int \frac{1-2x+4x^2}{x} dx = \int \frac{1}{x} + 2 + 4x dx$

$$= \ln(|x|) + 2x + 2x^2 + C$$

9. (8 points)

A drone is released into the air from an initial height of 5 feet off the ground. Its upward velocity at t seconds is given by $v(t) = \frac{1}{1+t^2}$ feet per second. (Recall: $\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$).

- a. How high will the drone be after 2 seconds? Please include units in your answer.

$$p(t) = \arctan(t) + C$$

$$= \arctan(t) + 5$$

$$p(2) = \arctan(2) + 5 \text{ ft}$$

after 2 seconds, the drone will be at $\arctan(2) + 5$ feet.

- b. Compute $\lim_{t \rightarrow \infty} v(t)$, and write a sentence which interprets your result in the context of the problem.

$$\lim_{t \rightarrow \infty} v(t) = \lim_{t \rightarrow \infty} \frac{1}{1+t^2} = 0$$

Note, $v(t) > 0$ for $t > 0$, so the drone is always going higher, but increasingly slower and practically plateauing at a height. Plainly: velocity is going to zero.

10. (6 points)

Suppose we want to estimate $\sqrt{4.1}$.

- a. Find the linear approximation (linearization) of the function $f(x) = \sqrt{x}$ when $a = 4$.

$$\begin{aligned} L(x) &= f(a) + f'(a)(x-a) \\ &= \sqrt{4} + \frac{1}{2}(4)^{-1/2}(x-4) \\ &= 2 + \frac{1}{4}(x-4) = 2 - \frac{1}{4}x + 1 = -\frac{1}{4}x + 3 \end{aligned}$$

- b. Use the linear approximation from part (a) to estimate $\sqrt{4.1}$. Your answer must be in the form of a simplified decimal or an exact fraction.

$$\begin{aligned} \sqrt{4.1} &\approx L(4.1) = -\frac{1}{4}(4.1) + 3 = 3 - \frac{4.1}{4} \\ &= 3 - \frac{41}{40} = \frac{120-41}{40} = \frac{79}{40} \end{aligned}$$

11. (Extra Credit: 5 points)

Compute $\lim_{x \rightarrow \infty} \sqrt{x^2-1} - x$. You must show your work.

Form $\infty - \infty$, try conjugate

$$\lim_{x \rightarrow \infty} \sqrt{x^2-1} - x = \lim_{x \rightarrow \infty} \frac{\sqrt{x^2-1} - x}{\sqrt{x^2-1} + x}$$

$$= \lim_{x \rightarrow \infty} \frac{(x^2-1) - x^2}{\sqrt{x^2-1} + x} = \lim_{x \rightarrow \infty} \frac{-1}{\sqrt{x^2-1} + x}$$

$$= \lim_{x \rightarrow \infty} \frac{-\frac{1}{x}}{\sqrt{1-\frac{1}{x^2}} + 1} = 0$$